



SHRI VAISHNAV VIDYAPEETH VISHWAVIDYALAYA

SHRI VAISHNAV INSTITUTE OF INFORMATION TECHNOLOGY,
INDORE



VOID HACKS () 8.0

Round 1: Idea Submission Round

Team name: Team_Cybernetic_100

S.no	Name	Email id	LinkedIn Link
1.	SUJAL PATIDAR	Sujalpatidar0511@gmail.com	www.linkedin.com/in/sujalpatidar05/
2.	SUJAL PATEL	sujalpatelt123@gmail.com	www.linkedin.com/in/sujal-patel-332087310/
3.	SHREYAS PUROHIT	shreyaspxb12@gmail.com	www.linkedin.com/in/shreyaspurohitdev/
4.	SUJEET RATHORE	sujeetrathore045@gmail.com	www.linkedin.com/in/sujeet-rathore045/

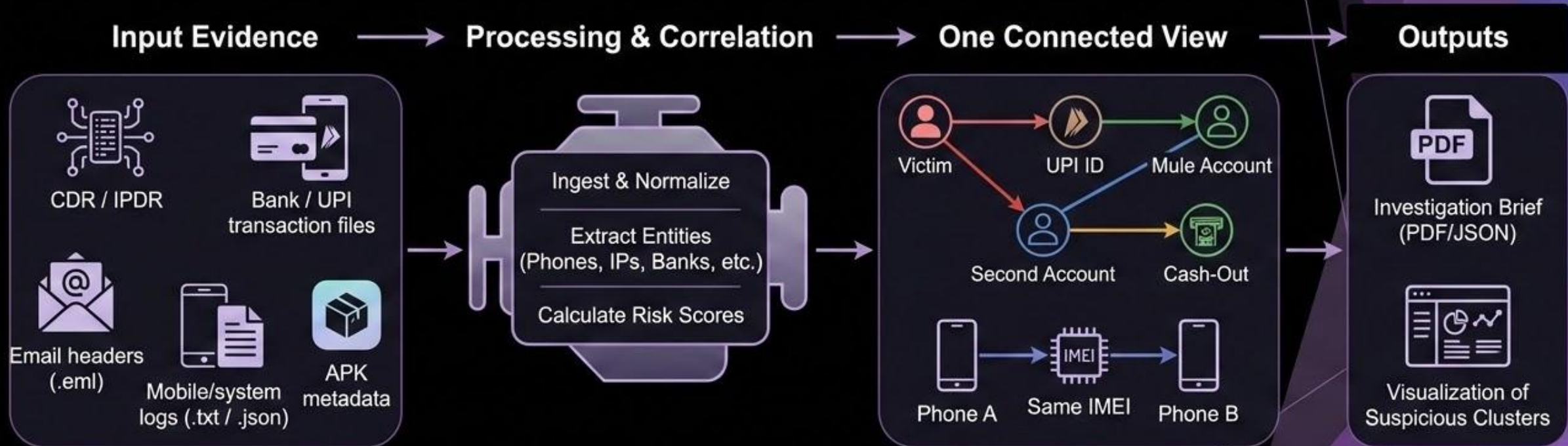
Team Leader Name: SUJEET RATHORE
Contact No. : 7804994722

IDEA TITLE



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❖ CyberTrace AI: AI-Powered Unified Cyber Fraud Analysis & Digital Artifact Correlator



AI does not independently declare a person guilty. It highlights suspicious relationships and patterns for the investigating officer.

TECHNICAL APPROACH



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Technologies to be Used



Programming Block
Python



Data Processing
Pandas



Graph Analysis
NetworkX



Visualization
PyVis / Plotly



Dashboard
Streamlit



Backend/API
FastAPI



Storage
SQLite



Evidence Integrity
hashlib /
SHA-256



AI Assistance
LLM / AI for
summarization
and explanation

Methodology and Process

1. Ingestion
Read CDR, IPOR,
bank/UPI, email, mobile
logs and APK metadata



2. Normalization
Convert different formats into
a common structure



3. Entity Extraction
Extract phone numbers,
IMEI, IMSI, UPI IDs,
accounts, IPs, MACs etc.



4. Entity Correlation
Find common identifiers and
relationships across datasets



5. Graph Creation
Create nodes and edges
representing people,
devices, accounts and
transactions



6. Risk Analysis
Detect suspicious patterns such as:

- Rapid multi-hop fund routing
- SIM switching
- Multiple accounts connected to one identifier
- Suspicious/spoofed email headers
- Unusual transaction patterns

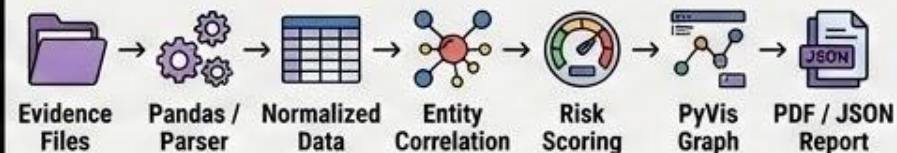


7. Report Generation
Generate a one-page PDF/JSON brief
containing:

- Suspects / entities
- Linked clusters
- Risk score
- Important relationships
- Recommended investigation/seizure leads



VISUAL DIAGRAM (PIPELINE)



Technology → NetworkX Stages

****Working Prototype****

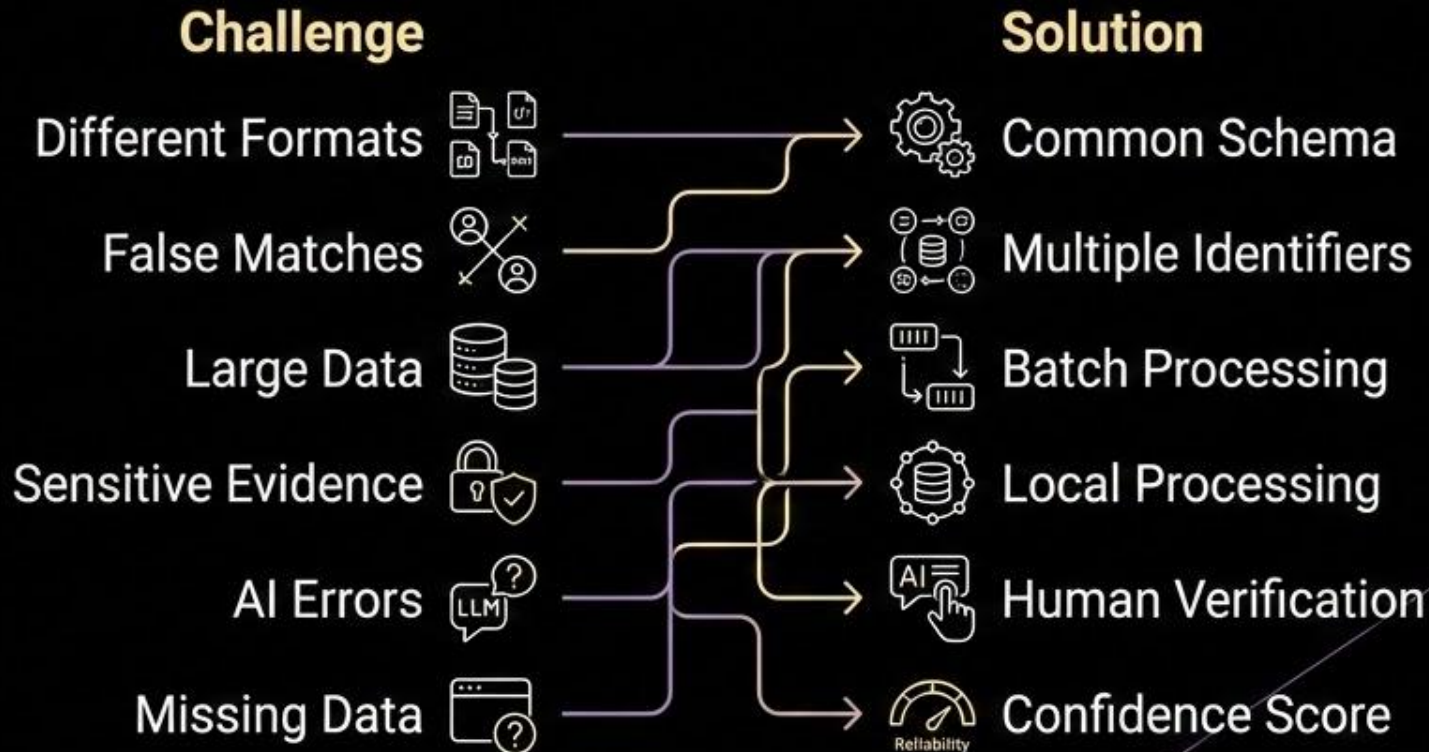
The PoC should demonstrate:
Upload Evidence → Process → Correlate →
Graph → Risk Score → Generate Report



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FEASIBILITY AND VIABILITY

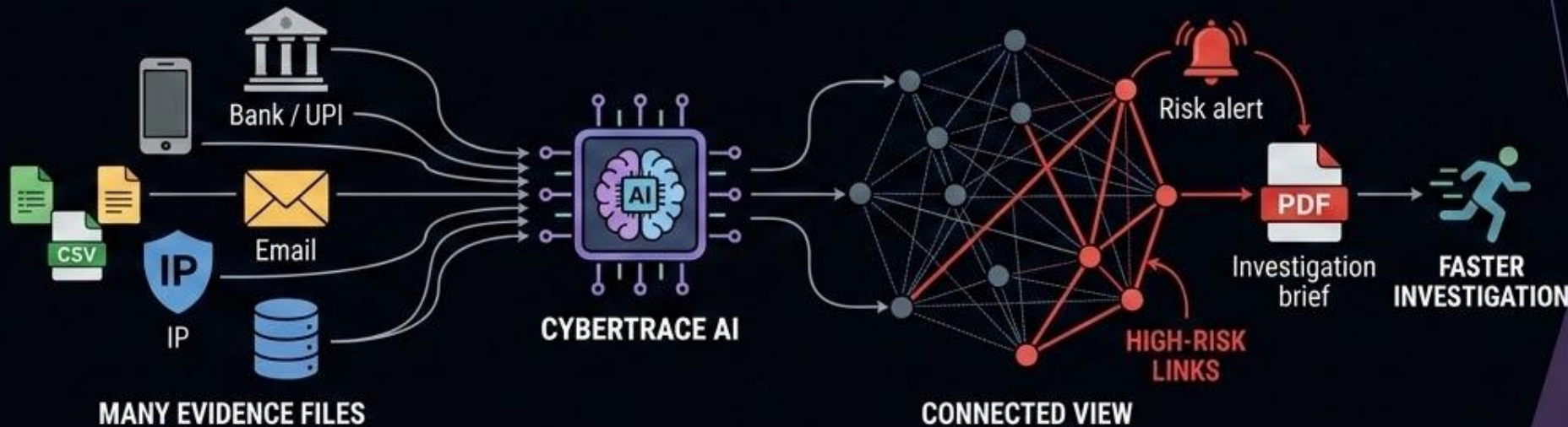
-  Uses open-source Python, common computers, and lightweight SQLite.
-  Can analyze data locally using mock datasets.
-  Keeps evidence offline for security.
-  Scalable for larger datasets later.





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IMPACT AND BENEFITS



Social Benefits



- FASTER INVESTIGATION
- CONNECT FRAGMENTED EVIDENCE

Economic Benefits



- REDUCE MANUAL EFFORT
- IDENTIFY MULE ACCOUNTS

Technical Benefits



- UNIFIED EVIDENCE ANALYSIS
- RISK-BASED TRIAGE

Connect the evidence. Find the hidden links. Prioritize the risk.

RESEARCH AND REFERENCES



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YEAR	RESEARCH WORK	PUBLISHER	WHY STUDIED	RELEVANCE
2022	Digital Forensics AI	Springer	To understand AI-assisted digital evidence mining and analysis	Digital Evidence Mining
2023	Detecting Anomalous Cryptocurrency Transactions	Springer	To study anomaly detection and suspicious transaction patterns	Fraud + Transaction Analysis
2023	Fraud Analytics: A Decade of Research	Elsevier	To understand major challenges and approaches in fraud analytics	Fraud Detection & Analytics
2024	Blockchain Forensics: Systematic Literature Review	MDPI	To study modern digital forensic techniques, evidence analysis and investigation challenges	Digital Forensics + Evidence
2025	Adaptive Metagraph Fraud Detection	Elsevier	To understand graph-based analysis for detecting hidden relationships in financial fraud	Graph-Based Fraud Detection

RESEARCH GAP

Existing studies focus on individual forensic and fraud-analysis tasks.

CyberTrace AI combines evidence correlation, graph analysis, risk scoring, and investigation reporting in one unified workflow.

Project Demo Video Link

- ▶ Participants may submit an accessible link to their project demo video hosted on YouTube, Google Drive, or any other platform.
- ▶ This is not mandatory, but it is part of the screening-round deliverables.
- ▶ Please ensure the link is accessible to the evaluation team.
- ▶ Link :-